

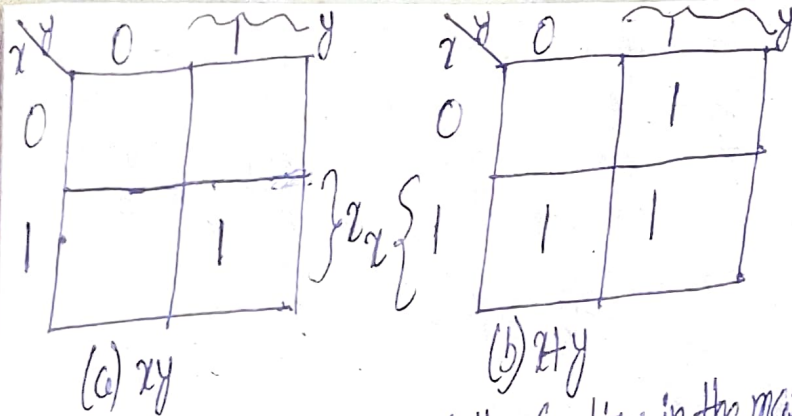


Figure 3-2 Representation of the functions in the map

A two-variable map is shown in Fig. 3-1. There are four minterms for two variables; hence the map consists of four squares, one for each minterm. As an example, the function xy is shown in Fig. 3-2(a). Since xy is equal to m_3 , a 1 is placed inside the square that belongs to m_3 . Since $x+y = x'y + xy' + xy = m_1 + m_2 + m_3$. The corresponding minterms are shown as 1 in Fig. 3-2(b).

Figure 3-3 Three-variable map

m_0	m_1	m_3	m_2
m_4	m_5	m_7	m_6

(a)

	y			
	00	01	11	10
xz	$x'y'z'$	$x'yz'$	$xy'z'$	xyz'
0	$x'y'z'$	$x'yz'$	$xy'z'$	xyz'
1	$x'y'z$	$x'yz$	$xy'z$	xyz

There are eight minterms for three binary variables. Therefore, a map consists of eight squares. The given sequence of arrangement of minterms has the following properties: Only one bit changes from 1 to 0 or from 0 to 1 in the listing sequence. To understand the usefulness of the map for simplifying Boolean functions, we must recognize the basic property possessed by adjacent squares. Any two adjacent squares in the map differ by only one variable which is primed in one square and unprimed in the other.